PUBLIC KEY CRYPTOGRAPHY

RSA Algorithm

RSA Algorithm

- Ron Rivest, Adi Shamir and Len Adleman have developed this algorithm (Rivest-Shamir-Adleman). It is a block cipher which converts plain text into cipher text and vice versa at receiver side.

- **The algorithm works as follow:**

1. Select two prime numbers p and q where $p \neq q$.
2. Calculate $n = p * q$.
3. Calculate $\Phi(n) = (p-1) * (q-1)$.
4. Select e such that, e is relatively prime to $\Phi(n)$
   i.e. $(e, \Phi(n)) = 1$ and $1 < e < \Phi(n)$
5. Calculate $d = e^{-1} \bmod \Phi(n)$ or $ed = 1 \bmod \Phi(n)$.
6. Public key = {e, n}, private key = {d, n}.
7. Find out cipher text using the formula,
   $C = P^e \bmod n$ where, $P < n$ and
   C = Cipher text, P = Plain text, e = Encryption key and n=block size.
8. $P = C^d \bmod n$. Plain text P can be obtain using the given formula.
   where, d = decryption key.

RSA algorithm
step by step
explanation with example

RSA Algorithm step by step explanation

- **Step – 1:** Select two prime numbers p and q where $p \neq q$.
- **Step – 2:** Calculate $n = p * q$.
- **Step – 3:** Calculate $\Phi(n) = (p-1) * (q-1)$.
- **Step – 4:** Select e such that, e is relatively prime to $\Phi(n)$

  i.e. $(e, \Phi(n)) = 1$ and $1 < e < \Phi(n)$

❖Explanation with example:

1. Two prime numbers $p = 13$, $q = 11$.
2. $n = p * q = 13 * 11 = 143$.
3. $\Phi(n) = (13 - 1) * (12 - 1) = 12 * 10 = 120$.
4. Select $e = 13$, $\gcd(13, 120) = 1$.

RSA Algorithm step by step explanation

➢ **Step – 5:** Calculate $d = e^{-1} \bmod \Phi(n)$ or $ed = 1 \bmod \Phi(n)$.

❖**Explanation with example:**

5. Finding d:

➔ $e * d \bmod \Phi(n) = 1$

➔ $13 * d \bmod 120 = 1$

(How to find: $d * e = 1 \bmod \Phi(n)$ ➔ $d = ((\Phi(n) * i) + 1) / e$

$d = (120 + 1) / 13 = 9.30$ ($\because i = 1$)

$d = (240 + 1) / 13 = 18.53$ ($\because i = 2$)

$d = (360 + 1) / 13 = 27.76$ ($\because i = 3$)

$d = (480 + 1) / 13 = 37$ ($\because i = 4$))

RSA Algorithm step by step explanation

- **Step – 6:** Public key = {e, n}, private key = {d, n}.
- **Step – 7:** Find out cipher text using the formula,

  $C = P^e \bmod n$ where, $P < n$

  C = Cipher text, P = Plain text, e = Encryption key and n=block size.
- **Step – 8:** $P = C^d \bmod n$. Plain text P can be obtain using the given formula.

  where, d = decryption key.

❖Explanation with example:

6. Public key = {13, 143} and private key = {37, 143}.

7. **Encryption :** Plain text ***P = 13***. (where, $P < n$)

   $C = P^e \bmod n = 13^{13} \bmod 143 = 52.$ ***C = 52***

8. **Decryption:**

   $P = C^d \bmod n = 52^{37} \bmod 143 = 13.$ ***P = 13***

Exercise - 1

Exercise – 1: P and Q are two prime numbers. P=7, and Q=17. Take public key E=5. If plain text value is 6, then what will be cipher text value according to RSA algorithm? Again calculate plain text value from cipher text.

Exercise - 2

Exercise – 2: In a public key cryptosystem using RSA algorithm, user uses two prime numbers 5 and 7. He chooses 11 as Encryption key, find out decryption key. What will be the cipher text, if the plaintext is 2? Decrypt the cipher text, what will be the value of plain text?